MILITARY SPECIFICATION

CONTROL UNITS
C-11628/ARC, C-10777A/ARC and C-11132/ARC
and
C-10320/ARC and C-10777/ARC

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, and test requirements for the C-11628/ARC, C-10777A/ARC, C-11132/ARC, C-10320/ARC and C-10777/ARC Control Units hereinafter referred to as the controls. The controls used in conjunction with a compatible Switching Unit (SA-2498/ARC or SA-2157/ARC) provide an automatic relaying mode (both plain and secure voice) when operating with two receiver-transmitter AN/ARC-182(V) or AN/ARC-159(V) (both same type or one of each), and one or two TSEC/KY-28, or TSEC/KY-58 speech security equipment(s).

1.2 Classification. Five types of controls are covered by this specification. The principle difference between the five controls (see 3.1) is the color of the panel lighting. The controls and respective lighting are as follows:

a.	C-11628/ARC	Navy Red Panel Lighting
b.	C-10777A/ARC	Navy White Panel Lighting
c.	C-11132/ARC	Air Force White Panel Lighting
d.	C-10320/ARC	Navy Red Panel Lighting
e.	C-10777/ARC	Navy White Panel Lighting

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems
MIL-W-5088	Wiring; Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-C-25050	Colors, Aeronautic Lights and Lighting Equipment, General Requirements for
MIL-M-38510	Microcircuits General Specification for
MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, Environmental Resistant, Removable Crimp Contacts; General Specification for

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MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-P-55110	Printed Wiring Boards

STANDARDS

Federal

FED-STD-595	Federal Standard Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	Electronic Equipments, General Requirements for
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 and Notice 4	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1472	Human and Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1562	Lists of Standard Microcircuits
MIL-STD-2076	Unit Under Test Compatibility with Automatic Test Equipment, General Requirements for
MIL-STD-2084	Maintainability of Avionics and Electronic Systems and Equipment, General Requirements for

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2.1.2 Other Government documents. The following other Government documents, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DOCUMENTS

National Security Agency

CSEEB-13	TSEC/KY-28 Speech Security Equipment Engineering Bulletin
CSEEB-14	Communications Secuity Equipment System Document for TSEC/KY-57/58
CSEEB-32	TSEC/KY-57/58 Communications Security Equipment Engineering Bulletin

Naval Air Systems Command

WS-6536	Process Specificatin Procedures and Requirements for Preparations and Soldering of Electrical Connections

Department of the Navy

NAVMAT P-4855.1	Navy Power Supply Reliability
NAVMAT P-9492	Navy Manufacturing Screening Program

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. The controls (C-11628/ARC, C-10777A/ARC, and C-11132/ARC) shall be self-contained units capable of controlling the SA-2498/ARC switching unit, and one or two TSEC/KY-28 or TSEC/KY-58 units from a remote location in all the modes specified in 3.6. The controls (C-10320ARC and C-110777/ARC) shall be self-contained units capable of controlling the SA-2157/ARAC switching unit, and one or two TSEC/KR-28 or TSEC/KY-58 units from a remote location in all the modes specified in 3.6.

3.2 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

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3.3 Parts. In the selection of parts, the prime considerations are fulfillment of performance requirements, logistics supportability and reliability assurance in a cost effective design. The following shall govern in parts selection.

a. Subassemblies shall be designed and fabricated to be repairable. Maximum economic standardization of parts shall be exercised. Parts selection shall be made in the following order of precedence:

(1) Military Standard Parts shall be selected in accordance with MIL-STD-454, Requirement 22.

(2) Nonstandard parts previously qualified for other programs.

(3) Nonstandard parts.

b. Nonstandard parts and nonestablished reliability military standard parts shall be screened to the requirements of the most similiar military standard or standard nonestablished reliability part. Integrated circuits and hybrids shall use MIL-STD-883, Method 5004, Class B, as a basis for tailored screens.

3.3.1 Component rescreening. The contractor shall conduct 100 percent incoming inspection of all active components. Functional and parametric checks of active components shall be performed at the component's rated temperature extremes and at ambient. Components which are not compliant with their specification shall not be used in the control.

3.3.2 Parts derating and application. All parts used shall be applied well within their ratings. The derating shall encompass the appropriate and meaningful application conditions such as voltage, current, power, temperature, mechanical, and duty cycle. Electronic and electromechanical parts shall conform to Table I electronic parts derating for worst case electrical and environmental stress unless formal written approval is received from the procuring activity prior to incorporation into the design. Part level stress analysis shall be used to verify that all parameter stresses are within the derated values at worst case circuit and environmental conditions. The following limitations or parts usage shall apply:

a. Prohibited application

(1) Non-hermetically sealed semiconductors except power transistors
(2) Photo couplers (photo transistors)
(3) Hot carrier (Schottky) rectifiers
(4) Microcircuit sockets
(5) Non-hermetically sealed wet tantalum capacitors

3.3.3 Parts control program. The contractor shall establish and maintain a parts control program in accordance with the requirements of MIL-STD-965, Procedure I and as specified herein. The program shall control part standardization and the criteria for the selection, application, and testing of parts to achieve the reliability required. As a minimum, a parts program equivalent to that specified herein shall be required of subcontractors and suppliers.

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Table I. Electronic parts derating.

Part Type	Parameter	Maximum Allowable Application Stress
Capacitor	Applied Voltage (Ripple Voltage for Tantalum	50% of rating 1/
Resistors	Power Voltage	50% of rating 70% of rating
Diodes	Power Current Voltage Junction Temperature	50% of rating 75% of rating 60% of rating 2/
Transistors	Power Voltage Junction Temperature	55% of rating 60% of rating 2/
Integrated Circuits		
Digital	Fan-Out, Fan-In Junction Temperature	80% of rating 2/
Linear	Power Junction Temperature	75% of rating 2/
Transformers, Inductive Devices	Parts shall be used at a winding temperature of at least 30 degrees below rated temperataure. In addition, the maximum voltage between windings or between a winding and a shield shall be not greater than 60% of rated voltage for the steady state conditions and 90% of rated voltage with transients. General purpose inductors, audio and power transformers shall be applied at no more than 70% of rated current.	

1/ For solid tantalum and all electrolytic capacitors, the maximum applied voltage shall be not greater than 70 percent of the vendor rated value. "Applied voltage" is the sum of the applied peak ripple voltage and the applied DC voltage.

2/ Semiconductor devices shall bemounted and operated in such manner to insure worst-case junction temperature is not greater than 110 degrees C at +71 degrees C ambient temperature. Any deviation requires procuring activity approval. Power semiconductor devices, on a very selective basis, may be mounted and operated with worst case junction temperatures not to be greater than 150 degrees C at +71 degrees C ambient temperatures.

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3.3.4 Non-standard parts approval. The contractor shall obtain procuring activity approval of all non-standard parts used in the equipment. Standard parts are defined in MIL-E-5400. In addition to the MIL-E-5400 requirements for standard parts, microelectronic parts are standard only if:

a. The device is listed in MIL-STD-1562, and

b. Sources (at least two) are listed on the Qualified Parts List (QPL), and

c. The part is procured from one of the qualified sources and

d. Quality level "B" (or better) of MIL-M-38510 is required.

The approval procedure shall be in accordance with MIL-E-5400 and MIL-STD-965, Procedure I. In addition to the requirements of MIL-E-5400, approval of non-standard parts will be based on:

a. Suitability for the application

b. Conformance to this specification and to MIL-E-5400

c. Interchangeability

d. Screening (burn-in) method

3.3.5 Samples of non-standard parts. Samples of non-standard parts are not required unless specified in the contract. Where non-standard parts are specified in the contract, the sample shall be in accordance with MIL-E-5400 (see 6.2.1).

3.3.6 Cabling and connections.

3.3.6.1 Cables and connectors. The controls shall provide for the use of cables and connectors in accordance with MIL-E-5400. The design of connectors shall be such that improper assembly, mating or installation is impossible. Non-RF connectors shall meet the requirements of MIL-C-38999. All external connectors shall be scoop-proof, have cadmium plating finish in accordance with QQ-P-416 (over a suitable underplating) to withstand a 500 hour salt spray test, and provide 360 degrees circumferential grounding techniques for grounding prior to contact engagement.

3.3.6.2 Interconnection cabling. The controls shall be capable of required operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provide the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the controls.

3.3.7 Control panels. The control panels shall conform to the applicable requirement of MIL-C-6781.

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3.3.7.1 Lighted panels-console mounted equipment. The plastic panels for all console mounted control panels, indicators, and radio sets shall conform to MIL-P-7788, Class 1-R, 1-W, or 1-BW Type V, with both 5- and 28-volt (ac and dc) lamp systems installed. Separate input power pins shall be provided for the two lighting systems. Color of panel lighting shall be in accordance with MIL-C-25050, as selected from the options by nomenclature assignment or by description on purchase orders listed in Table II.

Table II. Panel lighting color options

Nomenclature	Color (Short Name)	Color In Accordance With MIL-C-25050	MIL-P-7788 Class
C-11628/ARC	IPL Red	Type 1 (f)	1-R
C-10777A/ARC	IPL White	Type 1 (g)	1-W
C-11132/ARC	Blue Filtered White	Type 1 (h)	1-BW
C-10320/ARC	IPL Red	Type 1 (f)	1-R
C-10777/ARC	IPL White	Type 1 (g)	1-W

3.3.7.2 Console mounted equipment function marking. Console mounted control panels shall be permanently and clearly marked to indicate the function of the equipment with which they are associated.

3.3.7.3 Control knobs and functions. The number of controls required to operate the equipment shall be limited to and consistent with performing all the functions required by this specification. Closely associated controls such as control knobs on concentric shafts, shall be easily identified by contrasting colors of black and gray.

3.3.8 Interchangeability. Physical and functional interchangeability shall exist for all units and replaceable assemblies, subassemblies, and parts of the controls in accordance with MIL-E-5400. The controls shall be constructed to preclude the requirement for any adjustment when replacing an item by another item of the same type designation.

3.4 Design and construction. The control shall conform with all the applicable requirements of MIL-E-5400 for design and construction except where otherwise specified herein. In any case, it is a condition of final acceptance that the control shall meet all the performance requirements of this specification.

3.4.1 Reliability. Unless otherwise specified in the contract or purchase order, the contractor shall conduct a reliability program using MIL-STD-785 (see 6.2.1). Task 104 of MIL-STD-785 shall be implemented to establish and maintain a closed-loop failure reporting system. This system will identify failure trends and patterns at all levels of assembly prior to

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acceptance of the hardware and will also identify failure trends related to field failures throughout the contract period. The contractor shall develop and implement corrective action for significant failure trends. The reliability program shall include provisions for assuring the reliability of the control or any portion of the equipment supplied for use as spare or repair part.

3.4.1.1 Operational stability. The controls shall operate with required performance for at least the specified mean (operating) time between failures without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.4.1.2 Operating life. The controls shall have a total operating life of 20,000 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.4.1.3 Reliability in MTBF (mean time between failure). The equipment shall have a specified mean (operating) time between failures, upper test MTBF (θ_o) of 300,000 hours (as specified in MIL-STD-781), and a lower test MTBF (θ_1^o) of 150,000 hours.

3.4.2 Maintainability. The maintainability program, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in MIL-STD-2084.

3.4.2.1 Maintainability design. All electrical or electronic circuits and parts shall be packaged on replaceable and repairable, plug-in modules. Circuit breakers, indicators, and other items which may logically be bolted and soldered, or otherwise more rigidly electrically and mechanically fastened to control housing shall not be considered modules. SRAs (Shop Replaceable Assemblies) shall be QRA's (Quick Replaceable Assemblies) as specified in MIL-STD-2076. SRA arrangement shall be such that access to any SRA does not require the removal of adjacent SRAs or parts other than access panels. The use of a special tool or tools for SRA removal is prohibited. Fuses may be used in lieu of circuit breakers.

3.4.2.1.1 Keying. Connectors shall be keyed to prevent the insertion of a connector or subassembly into an improper location within or on the controls. All connectors shall be legibly marked with reference designations.

3.4.2.1.2 Access panels. Access panels shall be retained by fasteners suitable to the procuring activity.

3.4.2.1.3 Service access. Fasteners required to be loosened for removal of the WRA shall be held by retaining devices suitable to the procuring activity.

3.4.2.1.4 Adjustments. No adjustment or alignment shall be required at the organizational level. Adjustment or alignment if required at a higher maintenance level (intermediate or depot), shall not require the removal of the part to be adjusted or aligned, or of the hardware element to which it is attached.

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3.4.2.1.5 Test points. All test points shall be in accordance with MIL-STD-2084 and MIL-STD-415 to accommodate approved test equipment. MIL-STD-2084 takes precedence over MIL-STD-415. The sub-SRA test points and related connector requirements philosophy shall be reflected in appropriate maintenance data provided under the contract. Test points shall be provided for verification of faulty equipments at the intermediate level and repair of equipments and modules at the depot level. Conformal coating may be applied to the PC pads used as test points.

3.4.3 Transportability. The controls shall be transportable by air, rail, truck, and ship when packaged in accordance with the requirements of Section 5.

3.4.4 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests, except that required field tests may be conducted under outside ambient conditions:

a.	Temperature	Room ambient (25 $\pm$ 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative
e.	Input power voltage	28 $\pm$ 0.5 vdc

3.4.4.1 Service conditions. The controls shall operate as required under any of the environmental service conditions or combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.4.4.1.1 Sinusoidal vibration. The controls shall not require isolators or rear support and shall operate as required when subjected to sinusoidal vibration in accordance with test procedures of MIL-T-5422 to the requirements of Curve IIIB of MIL-E-5400.

3.4.4.1.2 Random vibration. The controls shall meet the random vibration requirements specified in 4.4.4.

3.4.4.1.3 Rain protection of console mounted equipment. The portion of the console mounted controls which are normally exposed when the equipment is mounted in a console shall meet the rain requirements of MIL-STD-810, Method 506.2, Procedure II.

3.4.4.1.4 Thermal conditions. The controls shall operate as required under the thermal conditions specified in MIL-E-5400, Class 2, without the requirement for auxiliary cooling.

3.4.4.1.5 Temperature shock. The control shall meet the temperature shock requirements of MIL-STD-810, Method 503.2.

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3.4.4.2 Warm-up time. The time required for the controls to warm up prior to operation shall be kept to a minimum and shall be not greater than 10 seconds at any operating environment.

3.4.4.3 Electromagnetic interference (EMI). The control unit shall comply with the requirements of MIL-STD-461 for Class A1b equipment except as modified by the specification. These requirements shall be met in all operating modes and at all operating frequencies of the control unit.

3.4.4.3.1 Conducted and radiated emissions limits. The conducted and radiated emissions of the control shall comply with the limits of the following requirements of MIL-STD--461, Part 2:

a. CE07

3.4.4.3.2 Grounding requirements. The grounding system between WRA's of the equipment shall be compatible and consistent with the following aircraft grounding scheme specified in 3.4.4.3.2.1 through 3.4.4.3.2.3. All WRA's shall be grounded in such a manner as to prevent ground loops and common ground returns for signal and power circuits, provide effective shielding for signal circuits, minimize EMI, and to protect personnel from electrical hazards.

3.4.4.3.2.1 Chassis grounds. A ground wire of minimum length connected internally to the WRA shcassis shall be provided at a pin on each primary power connector. No circuit shall be allowed to utilize this wire as its primary return.

3.4.4.3.2.2 Shielding grounding. A separate connector pin shall be provided for each aircraft wire shield unless suitable grounding is provided in external cable backshells. The connector shield pins shall be grounded to the equipment chassis inside the WRA adjacent to the connector mounting and by the shortest means practical.

3.4.4.3.2.3 Component grounds. All externally exposed metal parts, shields, control shafts, switch handles, connectors, and bushings, shall be grounded to the chassis.

3.4.4.3.3 Shielding gaskets. Shielding gaskets and similar devices for RF applications shall avoid the use of materials which represent severe dissimilar metal combinations with the interface.

3.4.5 Dimensions. The dimensions of the controls shall be as shown on Figure 1.

3.4.6 Weight. The weight of the control shall be not greater than 3.0 pounds.

3.4.7 Color. The front panel color of the control shall be lusterless black in accordance with FED-STD-595, Color No. 37038.

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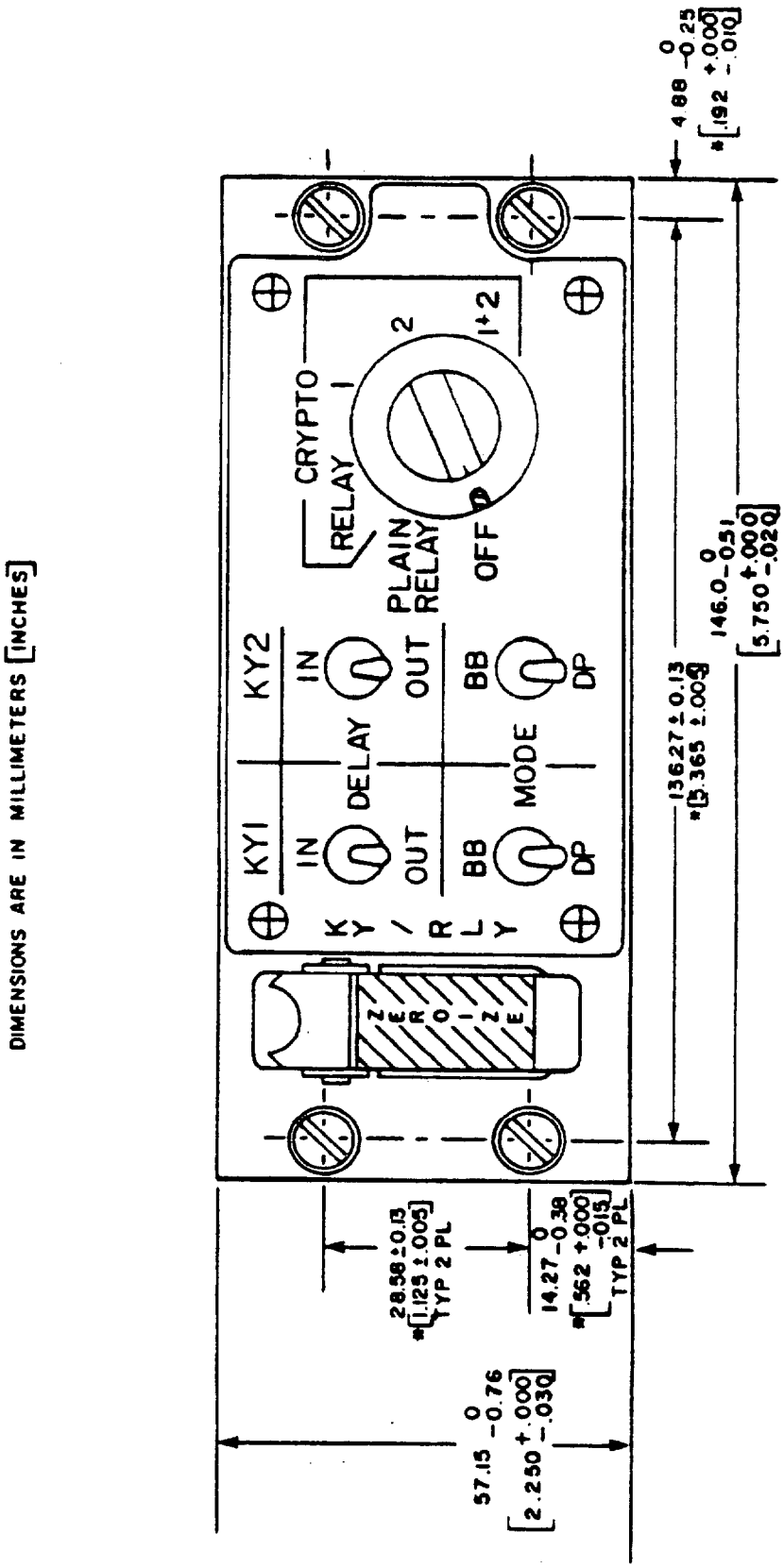


Figure 1. Outline dimensions for controls

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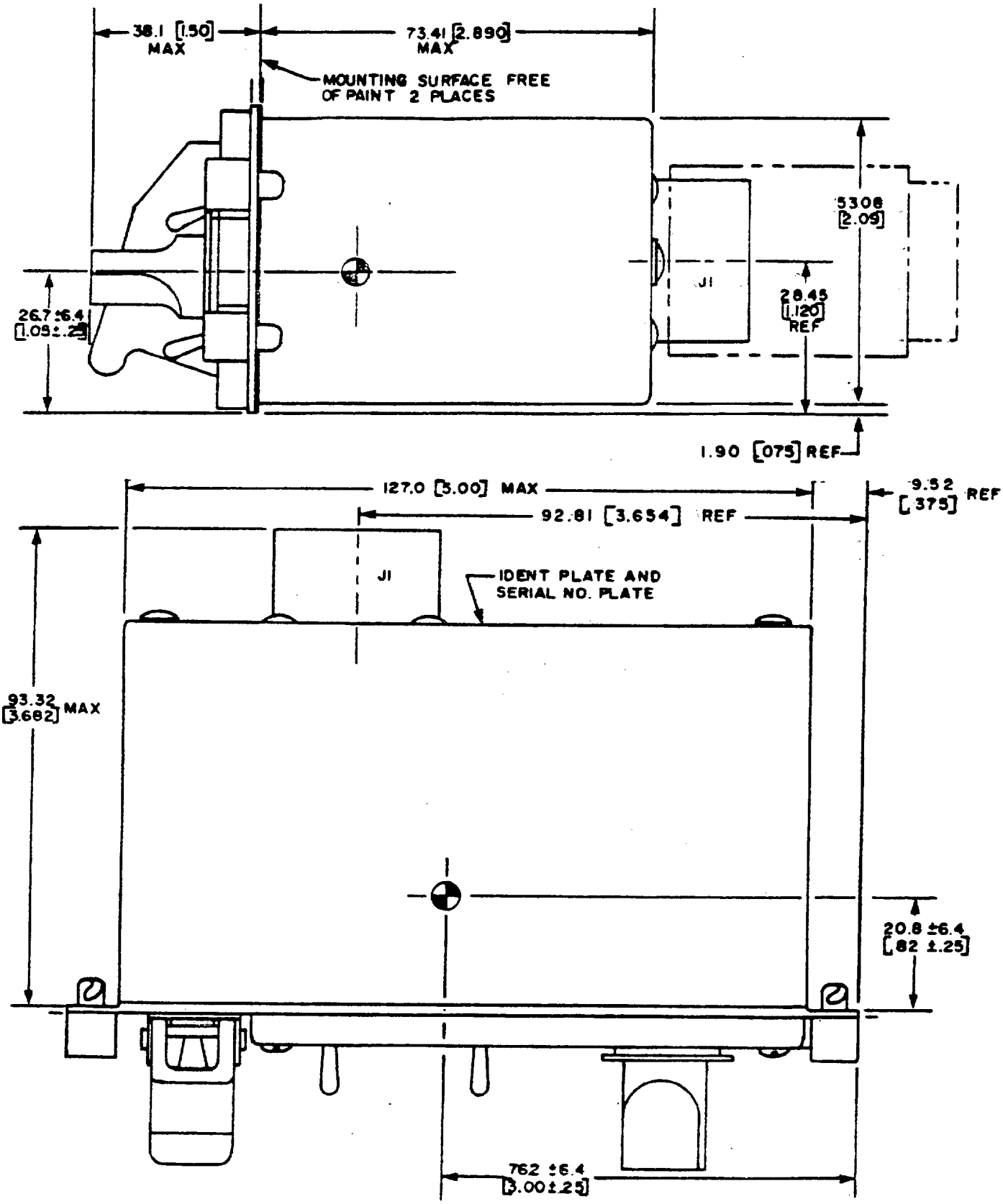


Figure 1. Outline dimensions for controls (Continued)

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3.4.8 Nameplates and identification marking. Nameplate approval, and identification marking shall be in accordance with MIL-N-18307.

3.4.8.1 Decals, nameplates and marking locations. All decals, nameplates, and marking shall be affixed to the controls as shown on Figure 1.

3.4.8.2 Serial numbers. Serial numbers will be assigned by the procuring activity for the controls under contract or purchase order. The contractor shall serialize sequentially all controls which are delivered under contract.

3.4.9 Input electrical power.

3.4.9.1 Operating power. The controls shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as specified in MIL-STD-704. The voltage supplied shall be 28 volts dc. The power required shall be not greater than 10 watts.

3.4.9.2 Lighting power. Input power for lighting of any lighted panel shall not require more than 5.0 watts at 5 or 28 volts (ac or dc).

3.4.9.3 Degraded performance. Degraded performance will be permitted for voltage transients specified by MIL-STD-704 with durations of 75 microseconds to 0.5 second during normal electrical system operation. Operations shall return to normal within five seconds with no resulting damage to the control.

3.4.9.4 Emergency voltage. During operation in the emergency steady state electric power condition specified in MIL-STD-704, the control shall remain safe, and control functions shall be maintained. The emergency power source will impose an interval of zero voltage at time of loss of primary power for a maximum period of 30 seconds, but shall return to performance as specified above after return to emergency power conditions in accordance with MIL-STD-704.

3.4.9.5 Undervoltage protection. The controls shall not be damaged by voltages below the minimums specified by MIL-STD-704 and shall resume normal operation when the voltage returns within normal limits.

3.4.9.6 Reverse polarity protection. The control shall be protected such that it will not be damaged by application of reverse polarity voltages.

3.4.10 Manufacturing requirements. The controls shall be manufactured in accordance with the manufacturing processes, workmanship, and quality assurance provisons of MIL-STD-454, MIL-STD-275, MIL-P-55110, WS-6536, NAVMAT P-4855.1, and NAVMAT P-9492.

3.5 Performance. Unless otherwise specified herein, values set forth to establish the requirements of specified performance apply to performance under both standard and service and input power conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

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3.5.1 Operation. The control shall provide the functions and capabilities specified herein. Additionally, the control shall be compatible with associated equipment listed in 6.5 to provide operations specified in 3.6.

3.6 Detail requirements.

3.6.1 Function. The control shall provide remote control of the switching unit in all modes specified in 3.6.3.1

3.6.2. Contents. The control shall be one self-contained unit suitable for cockpit mounting. It shall contain all the circuits and control facilities necessary to control a compatible switching unit (SA-2498/ARC or SA-2157) and one or two TSEC/KY-28 or TSEC/KY 58 units as described in CSEEB-13, CSEEB-14 and CSEEB-32, respectively, from a remote location.

3.6.3 Controls. Operating controls required to operate the remote equipment shall be located on the front of the control unit.

3.6.3.1 Operation modes.

3.6.3.3.1 OFF. In this mode, power shall be removed from the remote units but remains on in the control to allow the TSEC/KY zeroize functions to operate.

3.6.3.1.2 Plain relay. In this mode, the switching unit shall automatically relay plain text signals by switching audio and control functions betwen two transceiver units.

3.6.3.1.3 Crypto relay. In this mode, the units shall automatically relay cipher text information, but may or may not allow the relay/aircraft personnel to monitor the relay link or to transmit (relay override) in cipher mode, dependent on whether a TSEC/KY is installed.

3.6.3.1.4 Crypto 1. In this non-relay mode a single TSEC/KY shall be connected to RT-1 for cipher operation. RT-2 operates in plain text only.

3.6.3.1.5 Crypto 2. In this non-relay mode a single TSEC/KY shall be connected to RT-2 for cipher operation. RT-1 operates in plain text only.

3.6.3.1.6 Crypto 1 and 2. In this non-relay mode both RT-1 and RT-2 shall operate in cipher text mode. If only one secure unit is utilized, this position shall be inoperative.

3.6.3.1.7 Mode switches. Switches TSEC/KY to either BB (Baseband) or DP (diphase) mode. Open circuit for BB, ground for DP.

3.6.3.1.8 Delay switches. Controls built in preamble delay in the TSEC/KY units by applying a ground.

3.6.3.1.9 Zeroize switch. This switch shall remove the coded information from any installed TSEC/KY units. It switches a 14 ampere surge for 15 milliseconds.

3.6.3.2 Electrical connections. Connections to external circuits shall be provided on connector J1 (MIL-C-38999, Series III) as specified in Table III.

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Table III.a. C-11628/ARC, C-10777A/ARC and C-11132/ARC Remote Control Interface Characteristics J1-Power/Control Connector

Connector J1	Pin Function	I/O	Characteristics
1	Power Ground	I	Power Return (Internally Grounded)
2	Zeroize To KY-1	O	+28V = Zeroize, Open = Normal Parallel With Pin 5
3	+28V Plain Relay (J1)	O	+28V = Plain Relay Mode. Parallel with Pin 40
4	X-Mode Aud Sel To Radio 1	O	GND = Crypto Modes
5	Zeroize To KY-2	O	Same As Pin 2. Parallel with Pin 2
6	Delay Select To KY-2	O	GND = KY-2 Delay Mode
7	Primary Power To KY-2	O	+28V To KY-2 When Using 2 KY's. Jumper To Pin 26 for Only 1 KY.
8	Primary Power To KY-1	O	+28V = Crypto (Power For KY-1) Modes (Relay, 1, and 1+2))
9	Ground		
10	Ground		
11	Ground		
12	MIC 1 PTT (J1)	O	Ground When Radio 1 To Be In Transmit Mode. Internally Connected To Pin 13 and 25
13	MIC 1 PTT (J2)	O	Ground When Radio 1 To Be In Transmit Mode. Internally Connected To Pins 12 and 25
14	RE-XMT Control From Radio 1 (To SA)	O	Squelch Ground From Radio Set When Receiving Plain Voice Signal. Internally Connected To Pin 46

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Table III.a. C-11628/ARC, C-10777A/ARC and C-11132/ARC Remote Control Interface Characteristics J1-Power/Control Connector - Continued

Connector J1	Pin Function	I/O	Characteristics
15	+28V Crypto 1 (J2)	O	+28V = Crypto Modes (1,1+2). Parallel With Pin 47
16	+28V Crypto 2 (J2)	O	+28V = Crypto Modes (2, 1+2). Parallel With Pin 67
17	PT/CT Control KY-1	O	KY-1, 28V = PT, GND = CT
18	PT/CT Control KY-2	O	KY-2, 28V = PT, GND = CT
19	Open (Reserved)	N/A	
20	WB/NB Sel To Radio 2	O	GND = Crypto Modes (Wideband Select for Radio 2)
21	PTT To Radio 2	O	Ground When Radio 2 To Be In Transmit Mode. Internally Connected To Pin 56
22	RE-XMT Control From Radio 2	I	Squelch Ground From Radio Set When Receiving Plain Voice Signal. Internally Connected To Pin 74
23	Spare		
24	Primary Power +28 VDC	I	+28 Volts dc In Accordance With MIL-STD-704
25	MIC 1 PTT	I	Ground When Radio 1 To Be In Transmit Mode. Internally Connected to Pins 12 and 13
26	System Configuration	N/A	Jumper To Pin 7 For 1 KY. Not Used For 2 KY's.
27	System Configuration	N/A	Jumper To Pin 31 If ARC-159 Is Radio 2
28	Open (Reserved)	N/A	

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Table III.a. C-11628/ARC, C-10777A/ARC and C-11132/ARC Remote Control Interface Characteristics J1-Power/Control Connector - Continued

Connector J1	Pin Function	I/O	Characteristics
29	PTT To KY-1	O	Ground When KY-1 To Be In Transmit Mode. Internally Connected To Pin 64
30	X-Mode Aud Sel To Radio 2	O	GND = Crypto Modes
31	System Configuration	N/A	Jumper To Pin 27 If ARC-159 Is Radio 2
32	System Configuration	N/A	Jumper To Pin 33 If ARC-159 Is Radio 1
33	System Configuration	N/A	Jumper To Pin 32 If ARC-159 Is Radio 1
34	Delay To KY-1	O	GND = KY-1 Delay Mode
35	BB/DP Sel To KY-1	O	GND = KY-1 DP Mode
36	System Configuration	N/A	Jumper To Pin 37 For 2 KY's or Jumper To Pin 38 for 1 KY
37	System Configuration	N/A	Jumper To Pin 36 For 2 KY's. Not Used For Only 1 KY.
38	System Configuration	N/A	Jumper To Pin 39 For 2 KY's Or Jumper To Pin 36 For 1 Ky
39	System Configuration	N/A	Jumper To Pin 38 For 2 KY's. Not Used For Only 1 KY
40	+28V Plain Relay (J2)	O	+28V = Plain Relay Mode. Parallel With Pin 3
41	PTT From KY-58 1	I	Ground When KY-58 Is In XMIT Mode. Internally Connected To Pins 62 and 69
42	Spare		
43	Spare		

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Table III.a. C-11628/ARC, C-10777A/ARC and C-11132/ARC Remote Control Interfact Characteristics J1-Power/Control Connector - Continued

Connector J1	Pin Function	I/O	Characteristics
44	WB/NB Sel To Radio 1	O	GND = Crypto Modes (Wideband Select For Radio 1)
45	PTT To Radio 1	O	Ground When Radio 1 To Be In XMT Mode. Internally Connected To Pin 79
46	RE-XMT Control From	I	Squelch Ground From Radio Set When Receiving Plain Voice Signal. Internally Connected to Pin 46.
47	+28V Crypto 1 (j1)	O	+28V = Crypto Modes (1, 1+2). Parallel with Pin 15.
48	MIC 2 PTT	I	Ground When Radio 2 To Be In Transmit Mode. InternallyConnected To Pins 61 and 75.
49	PNL Lamp Power +28 VDC	I	+28V (AC or DC) 5 Watts Maximum (Use Pin 49 or Pin 50 Not Both)
50	PNL Lamp Power +5 VDC	I	+5V (AC and DC) 5 Watts Maximum (Use Pin 49 or Pin 50 Not Both)
51	System Configuration	N/A	Jumper to Pin 53 if 2 KY's Are Used.
52	PTT to KY-2	O	Ground When KY-2 To Be In Transmit Mode. Internally Connected to Pin 77
53	System Configuration	N/A	Jumper To Pin 51 If 2 KY's Are Used
54	System Configuration	N/A	Jumper To Pin 55 If Only 1 KY Is Used

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Table III.a. C-11628/ARC, C-10777A/ARC and C-11132/ARC Remote Control Interface Characteristics J1-Power/Control Connector - Continued

Connector J1	Pin Function	I/O	Characteristics
55	BB/DP Sel To KY-2	O	GND = KY-2 DP Mode. If 2 KY's Are Used. Jumper To Pin 54 If Only 1 KY Is used
56	PTT To Radio 2 (To SA)	I	Ground When Radio 2 To Be In Transmit Mode. Internally Connected To Pin 21
57	+28V Crypto Relay (J2)	O	Switched +28 Volts DC In Crypto Relay Mode. Internally Connected to Pin 72
58	Spare		
59	Spare		
60	Spare		
61	MIC 2 PTT (J2)	O	Ground When Radio 2 To Be In Transmit Mode. Internally Connected To Pins 48 and 75
62	PTT From KY-58 1 (To SA)	O	Ground When KY-58 1 Is In XMT Mode. Internally Connected To Pins 41 and 69
63	Ground Plain/Off	I	GND = Plain Mode
64	PTT To KY-1 (To SA)	O	Ground When KY-1 To Be In Transmit Mode. Internally Connected To Pin 29
65	PNL Lamp Power Comm	I	Return For Panel Lamps
66	+28V Plain/Off	O	+28V = Plain Mode
67	+28V Crypto 2 (J1)	O	+28V = Crypto Modes (1, 1+2) Parallel With Pin 16

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Table III.a. C-11628/ARC, C-10777A/ARC and C-11132/ARC Remote Control Interface Characteristics J1-Power/Control Connector - Continued

Connector J1	Pin Function	I/O	Characteristics
68	+28V Crypto 1+2	O	+28V = Crypto 1+2 Mode Only
69	PTT From KY-58 1 (To SA)	O	Ground When KY-58 Is In XMT Mode. Internally Connected To Pins 41 and 62
70	Spare		
71	Spare		
72	+28V Crypto Relay (J1)	O	Switched 28 Volts DC In Crypto Relay Mode. Parallel With Pin 57
73	Spare		
74	RE-XMT Control From Radio 2 (To SA)	O	Squelch Ground From Radio Set When Receiving Plain Voice Signal
75	MIC 2 PTT (J1)	O	Ground When Radio 2 To Be In Transmit Mode. Internally Connected To Pins 48 and 61
76	BB/DP Sel for KY-1 (To SA)	O	GND = KY-1 DP Mode
77	PTT To KY-2 (To SA)	I	Ground When KY-2 To Be In Transmit Mode. Internally Connected To Pin 52
78	Spare		
79	PTT To Radio 1 (To SA)	I	Ground When Radio 1 To Be in Transmit Mode. Internally Connected To Pin 45

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Table III.b. C-10320/ARC and C-10777/ARC
Remote Control Interface Characteristics
J1 - Power Control Connector

Connector J1	Pin Function	I/O	Characteristics
1	Spare		
2	Spare		
3	Spare		
4	Spare		
5	KY-1 Zeroize	O	28 = Zeroize, Open = Normal (Parallel with pin 6)
6	KY-2 Zeroize	O	Same as pins 5 (Parallel with pin 5)
7	Input Power	I	+28 volts dc in accordance with MIL-STD-704, 10 watts maximum
8	Plain/Cipher Control	O	KY-1, 28V = Inhibit, Gnd = Enable
9	Spare		
10	System Configuration	N/A	Jumper to pin 11 for 2 KY's or jumper to pin 12 for 1 KY.
11	System Configuration	N/A	Jumper to pin 10 for 2 KY's. Not used for only 1 KY.
12	System Configuration	N/A	Jumper to pin 13 for 2 KY's or jumper to pin 10 for 1 KY.
13	System Configuration	N/A	Jumper to pin 12 for 2 KY's. Not used for only 1 KY.
14	Plain/Cipher Control	O	KY-2, 28V - Inhibit, Gnd = Enable
15	28V Crypto Relay	O	Switched 28 volts dc, 15 watt maximum
16	Spare		

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Table III.b. C-10320/ARC and C-10777/ARC
Remote Control Interface Characteristics
J1 - Power Control Connector - Continued

Connector J1	Pin Function	I/O	Characteristics
17	System Configuration	N/A	Jumper to pin 21 for 1 KY. Not used for 2 KY's.
18	28V Crypto 1	O	28V = Crypto Modes
19	28V Crypto 1	O	28V = Crypto (Power for KY-1)
20	28V Crypto 2	O	28V = Crypto Modes
21	System Configuration	O	28V to KY-2 when using 2 KY's. Jumper to pin 17 for only 1 KY.
22	28V Crypto 1 + 2	O	28V = Crypto 1 + 2 Modes only
23	KY Audio Select 1	O	Gnd = Crypto Modes
24	Spare		
25	Spare		
26	KY Audio Select 2	O	Gnd = Crypto Modes
27	Bandwidth Select 1	O	Gnd = Crypto Modes (Wideband select for Radio 1)
28	Bandwidth Select 2	O	Gnd = Crypto Modes (Wideband select for Radio 2)
29	28 V Panel Lamps	I	28V (AC or DC) 5 Watts maximum (use pin 29 or pin 31, not both)
30	Panel Lamps Common	I	Return for panel lamps
31	5V Panel Lamps	I	5V (AC or DC) 5 watts maximum (use pin 31 or pin 29, not both)
32	Spare		

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Table III.b. C-10320/ARC and C-10777/ARC
Remote Control Interface Characteristics
J1 - Power Control Connector - Continued

Connector J1	Pin Function	I/O	Characteristics
33	Power Ground	I	Power return (Internally grounded)
34	System Configuration	N/A	Jumper to pin 35 if ARC-159 is Radio 2
35	System Configuration	N/A	Jumper to pin 34 if ARC-159 is Radio 2
36	System Configuration	N/A	Jumper to pin 37 if ARC-159 is Radio 1
37	System Configuration	N/A	Jumper to pin 36 if ARC-159 is Radio 1
38	BB/DP Select 1	O	Gnd - KY-1 DP Mode
39	Spare		
40	Spare		
41	BB/DP Select 2	O	Gnd = KY-2 DP Mode if 2 KY's are used. Jumper to pin 42 if only 1 KY is used.
42	System Configuration	N/A	Jumper to pin 41 if only 1 KY is used
43	System Configuration	N/A	Jumper to pin 44 if only 2 KY's are used
44	System Configuration	N/A	Jumper to pin 43 if 2 KY's are used
45	Ground Plain/Off	O	Gnd = Plain Mode
46	Spare		
47	Spare		
48	28V Plain/Off	O	28V = Plain Mode
49	28V Plain Relay	O	28V = Plain Relay Mode

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Table III.b. C-10320/ARC and C-10777/ARC
Remote Control Interface Characteristics
J1 - Power Control Connector - Continued

Connector J1	Pin Function	I/O	Characteristics
50	KY-1 Delay	O	Gnd = KY-1 Delay Mode
51	KY-2 Delay	O	Gnd = KY-2 Delay Mode
52	Spare		
53	Spare		
54	Spare		
55	Spare		

3.7 Safety. The control safety requirements shall be in accordance with MIL-E-5400 except resistance from ground to chassis shall be not greater than 20 milliohms.

3.7.1 Human performance/human engineering. The requirements of MIL-STD-1472 and MIL-H-46855 shall be used as design guides for the controls and all components therein.

3.8 Bonding. Bonding shall be in accordance with MIL-B-5087.

3.9 Workmanship. Workmanship for the controls shall conform with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Standard conditions. Unless otherwise specified herein, all inspections shall be performed with conditions specified as follows:

a.	Temperature	Room ambient (25 $\pm$ 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power	28 $\pm$ 0.5 volt dc

4.4 First article inspection. First article inspection shall be conducted by the contractor on controls representative of the production control to be supplied under the contract. First article tests shall be accomplished under the authorized test procedures (see 4.6). No first article inspections shall be conducted prior to acceptance of the first article test procedure by the procuring activity. First article inspection shall consist of all the tests described in 4.4.1.

4.4.1 Scope of first article tests. First article tests shall include all tests in the approved test procedure to determine that the control meets all the requirements of this specification, other applicable specifications, and the contract. First article tests shall include environmental tests in accordance with the procedures of MIL-T-5422 including temperature shock, a random vibration test as specified in 4.4.4, and an electromagnetic compatibility test in accordance with MIL-STD-461 and MIL-STD-462, as modified in 4.4.3. The first article test may also include a reliability qualification test.

4.4.2 First article approval. Approval of the first article sample will be by the procuring activity upon completion of all tests and the first article test report being prepared for the procuring activity.

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4.4.3 Electromagnetic interference test. Compliance with the requirements of 3.4.4.3 shall be demonstrated by tests. These tests shall be performed in accordance with MIL-STD-462 with Notice 4 as modified herein.

4.4.3.1 EMI Test Plan. The contractor shall perform EMI tests in accordance with an EMI Test Plan. The Test Plan shall describe all EMI tests required to demonstrate compliance with the requirements of 3.4.4.3, and shall addresss all anomalies, inconsistencies, and omissions (if any). The Test Plan shall detail a combination of operating modes and frequencies to be tested. The combination shall represent all operating modes of the control. The Test Plan shall be acceptable to the procuring activity prior to the commencement of EMI testing.

4.4.3.2 CE07 test procedure. The contractor shall prepare a test to demonstrate compliance with the CE07 requirement of 3.4.4.3.1. This test shall be included as a part of the EMI Test Plan.

4.4.4 Random vibration test. A random vibration test shall be conducted as specified in 4.4.4.1 through 4.4.4.4.

4.4.4.1 Test item operation. The test item shall be operated during application of random vibration so that the functional effects caused by these tests may be evaluated. The test item shall meet performance requirements, as specified, while the functional vibration levels are being applied and immediately preceding and following the application of the endurance levels.

4.4.4.2 Mounting techniques. The test item shall be hard mounted either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of extraneous responses in the test setup. The test load shall be distributed as uniformly as possible on the vibration exciter table in order to minimize effects of unbalanced loads. The input control sensing device(s) shall be rigidly attached to the vibration table, or fixture if used, as near as possible to the attachment point(s) of the test item. Additional vibration sensors shall be located in or on the test item to determine resonant frequencies and amplification factors. Locations to be selected should include main structure, printed circuit boards, large components, and modules, where practicable. The sensor sizes and weights shall be limited so that their effect on the dynamic responses being measured is minimal.

4.4.4.3 Performance of test. The individual equipment test item shall be subjected to broadband random vibration excitation with the power spectral density envelope shown on Figure 2. The test item shall be attached to the vibration exciter according to 4.4.4.2. Vibration shall be applied sequentially along each of the three orthogonal axes of the test item. Two test levels are required, a functional level and an endurance level. For each axis, one half of the functional test shall be conducted first, then the endurance test, followed by the second half of the functional test. The equipment shall perform according to the operating requirements as specified in 4.4.4.1. The acceleration power spectral density (g^2/Hz) of applied vibration, as measured on the test fixture at mounting points of the test item, shall be as specified in 4.4.4.4. Test times shall, for each axis, be

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one hour each for functional and endurance levels. The instantaneous random vibration acceleration peaks may be limited to three times the rms acceleration level. The power spectral density of the test control signal shall not deviate from the specified requirements by more than +100, -30 percent (+3, -1.5 dB) below 500 Hz and +100, -50 percent (+3 dB) between 500 Hz and 2,000 Hz except that deviations as large as +300, -75 percent (+6 dB) shall be allowed over a cumulative bandwidth of 100 Hz maximum, between 500 and 2,000 Hz. Tolerance levels in terms of dB are specified as:

$$dB = 10 \text{ Log}_{10} \frac{W1}{W0}$$

Where W1 = measured acceleration power spectral density in g^2/Hz units, the term W0 defines the specified level in g^2/Hz units. Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant product, BT=50 minimum. Specific analyzer characteristics shall be as specified below or equivalent, subject to the BT=50, minimum limitation.

a. On-line, contiguous filter, equalization analysis system having a bandwidth as follows:

B=25 Hz, maximum between 20 and 200 Hz
B=50 Hz, maximum between 200 and 1,000 Hz
B=100 Hz, maximum between 1,000 and 2,000 Hz

b. Swept frequency analysis systems characterized as follows:

(1) Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B=25 Hz, maximum between 20 to 200 Hz
B=50 Hz, maximum between 200 to 1,000 Hz
B=100 Hz, maximum between 1,000 to 2,000 Hz

(b) Analyzer averaging time =T= 2 RC = 1 second, where T= true averaging time constant

(c) Analysis sweep rate (linear) =

$$R = \frac{B}{4RC} \text{ or } \frac{B^2}{8} \text{ (Hz/second)}$$

(2) Constant percentage bandwidth analyzer

(a) Filter bandwidth - pfc=one-tenth of center frequency maximum (0.1 fc) where p=percentage and fc=analyzer center frequency

(b) Analyzer average time = T = $\frac{50}{pfc}$, minimum

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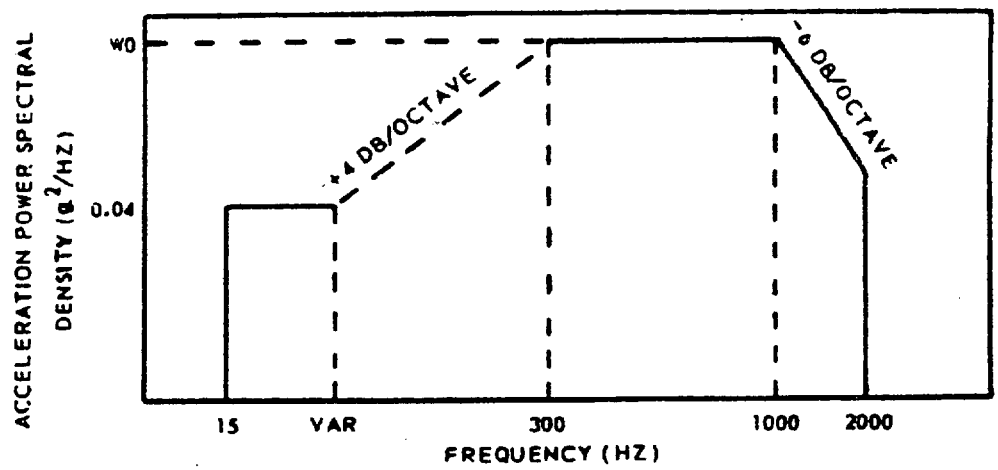


FIGURE 2. Random vibration envelope

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(c) Analysis sweep rate (logarithmic) =

$$T = \frac{pfc}{4RC} \text{ or } \frac{(pfc)^2}{8}$$

(Hz/second), maximum, whichever is smaller

Digital power spectral density analysis employing quantization techniques providing accuracies corresponding to the above approach.

4.4.4.3.1 Worst case axis. During the vibration testing, data shall be obtained to identify the worst case axis for purpose of vibration screening tests on production equipments.

4.4.4.4 Test levels. For the purpose of the random vibration test, the following test levels shall be used:

a. Functional Level-W_o = 0.05 g^2/Hz (8.5g rms minimum)

b. Endurance Level-W_o = 0.12 g^2/Hz (12.7g rms minimum)

4.4.5 Production controls. Controls supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be capable of successfully passing the same tests as specified herein.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests. All inspection and testing may be under the supervision of the procuring activity representative. The contractor shall retain test data showing quantitative results for all quality conformance tests. Such tests shall be signed and stamped by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the ifnished product. Quality confromance tests shall consist of the following:

a. Individual tests (see 4.5.1)

b. Sampling tests (see 4.5.3)

c. Special tests (see 4.5.4)

4.5.1 Individual tests. Each control sumitted for acceptance shall be subjected to the individual tests. The tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliabiilty. As a miinimum, each control accepted shall have passed the following tests:

a. Examination of produce (see 4.5.1.1)

b. Operational test (see 4.5.1.2)

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4.5.1.1 Examination of product. Each control shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each control shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure specified equipment operation.

4.5.2 Sampling tests. Controls shall be subjected to sampling tests. Controls selected for sampling tests shall first have passed the individual tests. Controls will be selected for sampling tests by the government inspector in accordance with Table IV.

Table IV. Sampling test sample selection

Quantity of equipments offered for acceptance	Quantity to be selected for sampling test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling tests are not required when reliability assurance tests are conducted.

4.5.3.1 Scope of sampling tests. As a minimum, each control selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each control tested and should be based on results of the first article, individual and special tests.

4.5.4 Special tests. When specified in the contract or purchase order (see 6.2.1), special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the control and to assure quality control. The control selected for special tests may be selected from controls previously subjected to the sampling or reliability tests.

4.5.4.1 Special test schedule. Selection of controls for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

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b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.5.4.2 Scope of special tests. Special tests shall consist of such tests as authorized by the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and make it available to the procuring activity for acceptance prior to conducting the tests.

4.6 Test procedures. The procedures used for conducting all tests required by Section 4 shall be prepared by the contractor and made available to the procuring activity. The right is reserved by the procuring activity or the government representative to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of the specification or the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of test procedures.

4.7 Presubmission testing. No item, part, or complete equipment shall be submitted until it has been previously tested by the manufacturer and found to comply, with all applicable requirements of this specification and the contract.

5. PACKAGING

5.1 General. All major units and parts of the controls shall be preserved, packaged, packed, and marked in accordance with MIL-E-17555 for the level of shipment specified in the contract or purchase order (see 6.2.1).

5.2 Marking. Shipments shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The controls specified herein are used to control the switching of various audio signals within a compatible Switching Unit and to control the operation of one or two TSEC/KY units (see 3.1).

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification

b. If a first article inspection is required (see 3.2, 6.4)

c. If samples of non-standard parts are required (see 3.2, 6.4)

d. If reliability program is not to be established by contractor (see 3.4.1)

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e. If Special Tests are required (see 4.5.5)

f. Test procedure modifications or additional tests, if required (see 4.6)

g. Selection of applicable levels of packaging and packing (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement title	Applicable DID no.
3.3.2	Engineering Change Proposals (ECPs) and Requests for Deviations and Waivers (Long Form)	DI-E-2037
3.3.3	Program Parts Selection List (PPSL)	DI-MISC-80072
3.4.1	Plan, Failure Data Collection Analysis and Corrective Action	UDI-T-23719
3.4.1.2	Parts Approval Request	DI-MISC-80071
3.4.8	Request for Confirmation of Nomenclature	DI-CMAN-80194
	Request for Assignment of Serial Number and Serial Number Prefix Letters	DI-CMAN-80195
	Request, Assignment of Nomenclature	UDI-E-21582
	Request for Approval of Identification Plate Drawings	DI-CMAN-80196
4.4, 4.6	First Article Inspection Procedure	DI-T-4901
4.4.2	First Article Inspection Report	DI-T-4902
4.4.3.1, 4.4.3.2	Electromagnetic Interference Test Plan	DI-EMCS-80201
4.5.4.2	Production/Acceptance Inspection Procedures	DI-T-4903

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.

6.3 Options. An additional burn-in period is allowable at the option of the contractor.

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to rwaive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, may furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Associated equipment. These controls should operate with the associated equipment listed in Table V which should not be supplied as part of this equipment:

Table V. Associated equipment

Item	Equipment Designation	Military Specification
Speech Security Equipment	TSEC/KY-28 TSEC/KY-58	CSEEB-13 CSEEB-32/CSESD-14
Receiver-Transmitter, Radio	RT-1324A/ARC RT-1327A/ARC RT-1407/ARC	MIL-R-85665 MIL-R-85665 MIL-R-85665
Switching Unit	SA-2498/ARC	MIL-S-85673
Receiver-Transmitter	RT-1250A/ARC	MIL-R-85664

6.6 Subject term (key word) listing

AN/ARC-182 (v)
Cipher text
Control head
Crypto
Relay
Switching Unit Control
TSEC/KY-28
TSEC/KY-58

Preparing Activity:
Navy AS
(Project 5821-N198)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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